Course Title: Ethical Hacking

Objective

The objective of this experiment is to perform Open Source Intelligence (OSINT) and passive reconnaissance using FOCA, Google Dorking, and publicly available online resources. These techniques help gather metadata, exposed documents, domain information, and other publicly accessible information to understand an organization's digital footprint without directly interacting with its systems.

Methodology

1. Install and configure FOCA for metadata extraction and OSINT analysis.
2. Use Google Dorking techniques to discover publicly indexed files and exposed information.
3. Use FOCA to collect metadata from publicly available documents and identify hosts, domains, emails, and software information.
4. Perform passive reconnaissance using public websites and search operators without directly interacting with the target system.
5. Analyze the collected information to understand the organization's digital footprint and identify potential security exposures.

Expected Outcome

By completing this experiment, you will be able to:

1) Install and configure the **FOCA** tool successfully.
2) Perform **Google Dorking** to discover publicly available information.
3) Extract document metadata using **FOCA**.
4) Identify domains, hosts, email addresses, and document information through passive reconnaissance.
5) Gain practical experience with **OSINT** and **footprinting** techniques.
6) Improve the ability to analyze publicly available information for cybersecurity assessments and reconnaissance.

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1. Introduction to FOCA

FOCA (Fingerprinting Organizations with Collected Archives) is an **Open Source Intelligence (OSINT)** tool used to discover and analyze publicly available documents from websites. It extracts **metadata** from files such as PDF, Word, Excel, and PowerPoint documents to identify valuable information about an organization.

Features

- Discovers publicly available documents.
- Extracts document metadata.
- Identifies domains, subdomains, and IP addresses.
- Finds email addresses and usernames.
- Supports passive reconnaissance and footprinting.

Applications

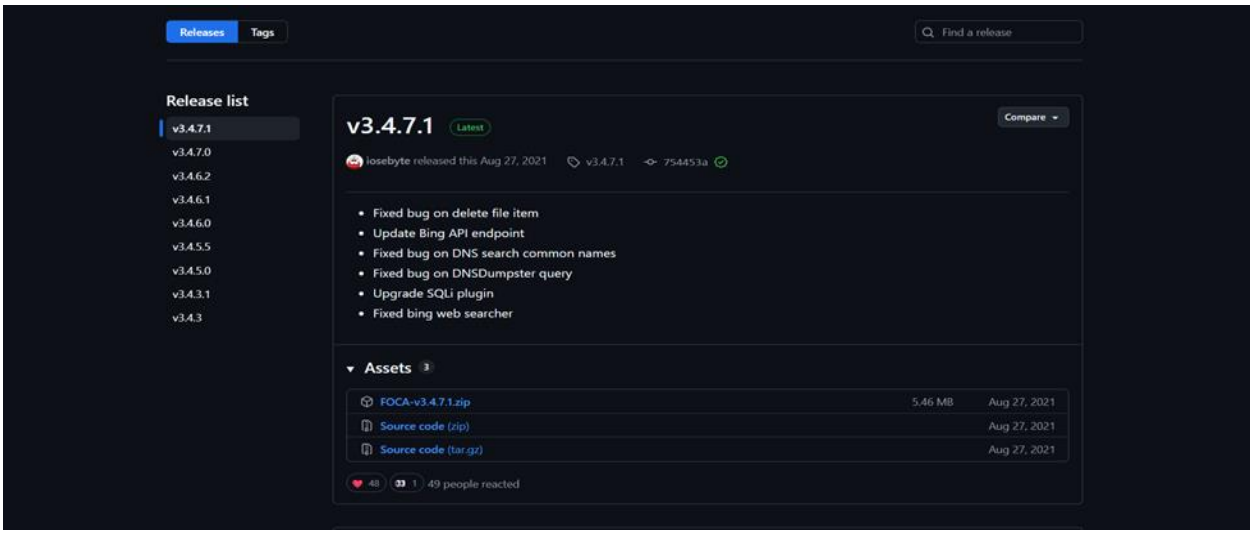
- OSINT (Open Source Intelligence)
- Footprinting
- Passive Reconnaissance
- Security Assessment
- Digital Footprint Analysis

2. Installing FOCA

Step 1: Download FOCA

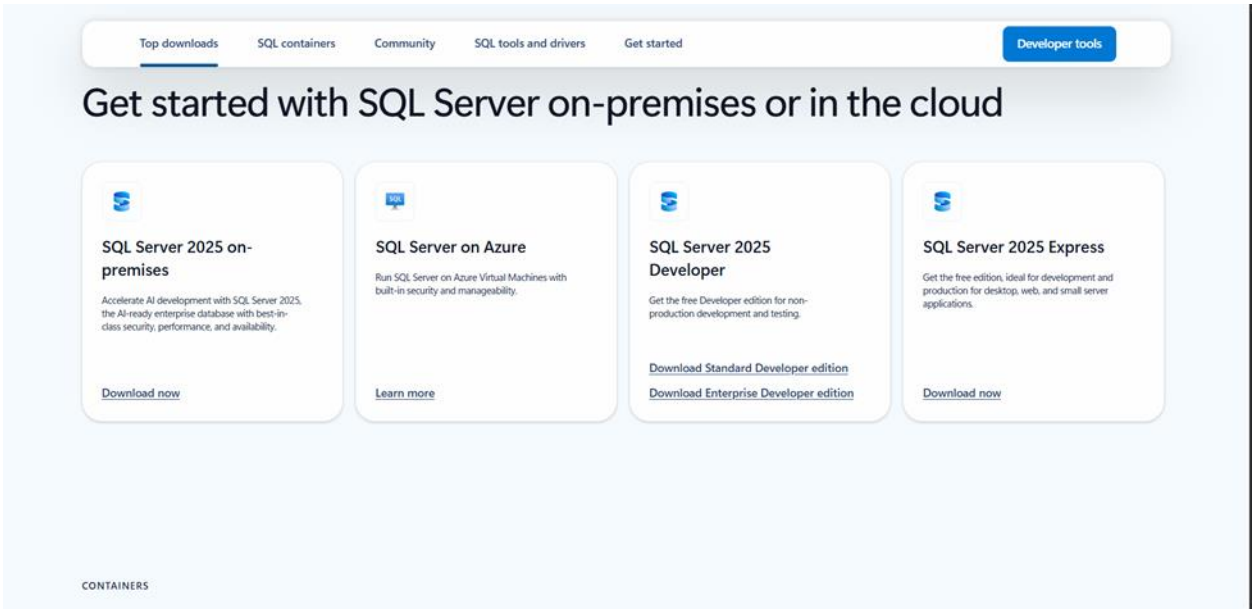
Download the latest version of **FOCA** from the official GitHub Releases page.

https://github.com/ElevenPaths/FOCA/releases



Step 2: Download SQL Server Express

Download and install **Microsoft SQL Server Express**, which is required for FOCA.
https://www.microsoft.com/en-us/sql-server/sql-server-downloads

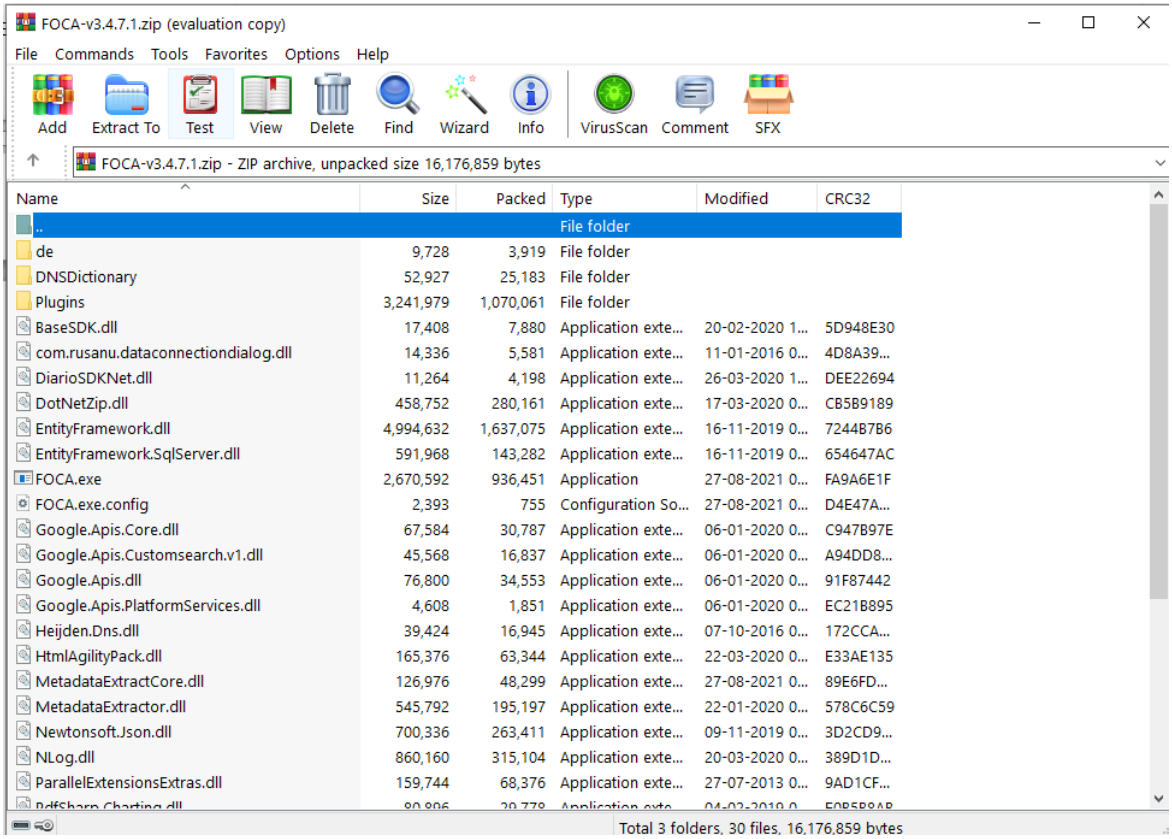


Step 3: Install SQL Server

Open the SQL Server installer, select **Basic**, accept the license agreement, and complete the installation.

Step 4: Launch FOCA

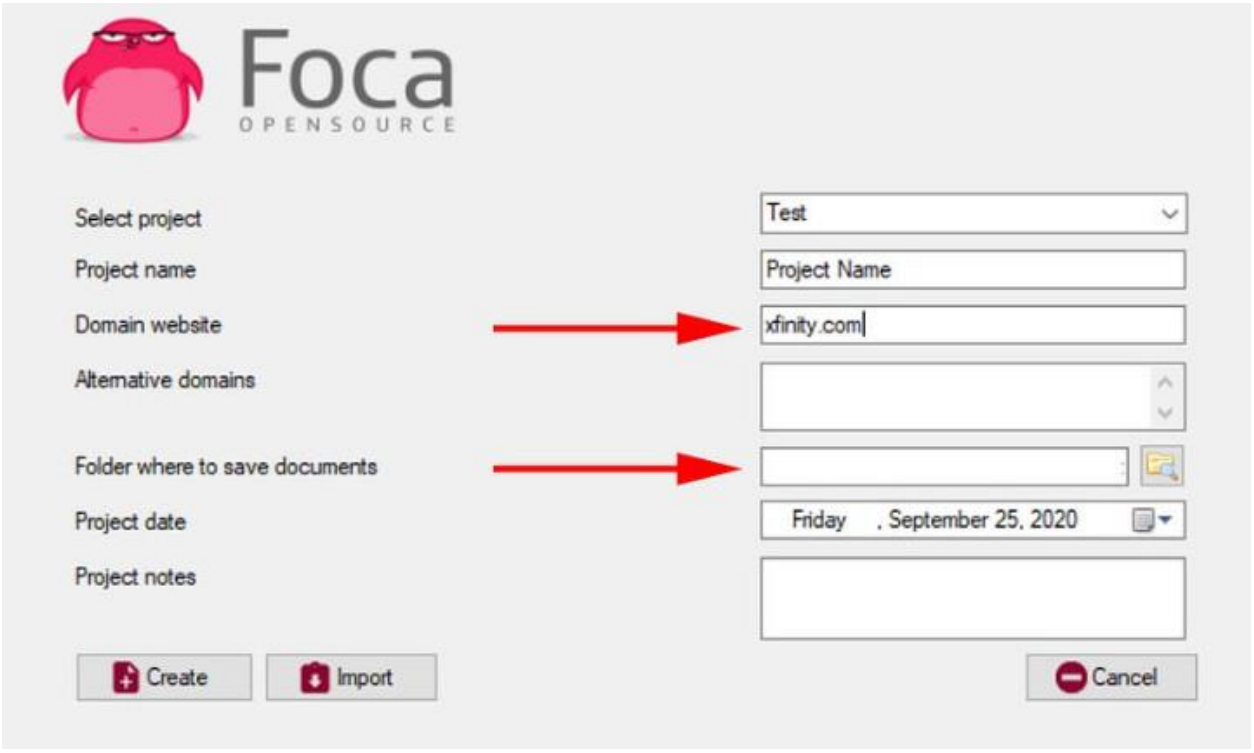
Extract the downloaded FOCA ZIP file and run **FOCA.exe**.

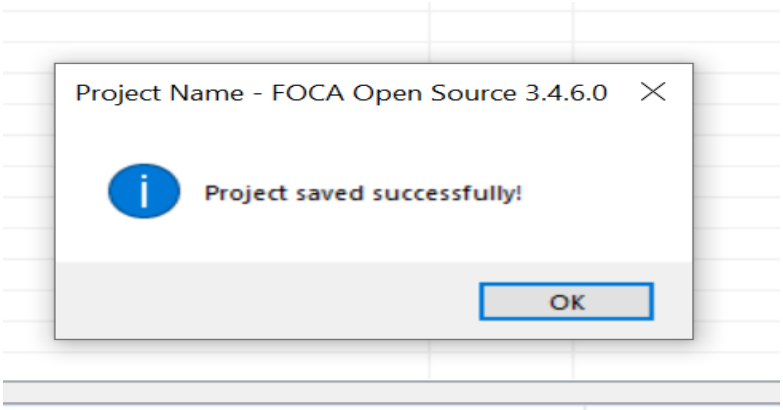


3. Creating a New Project

Step 5: Create a New Project

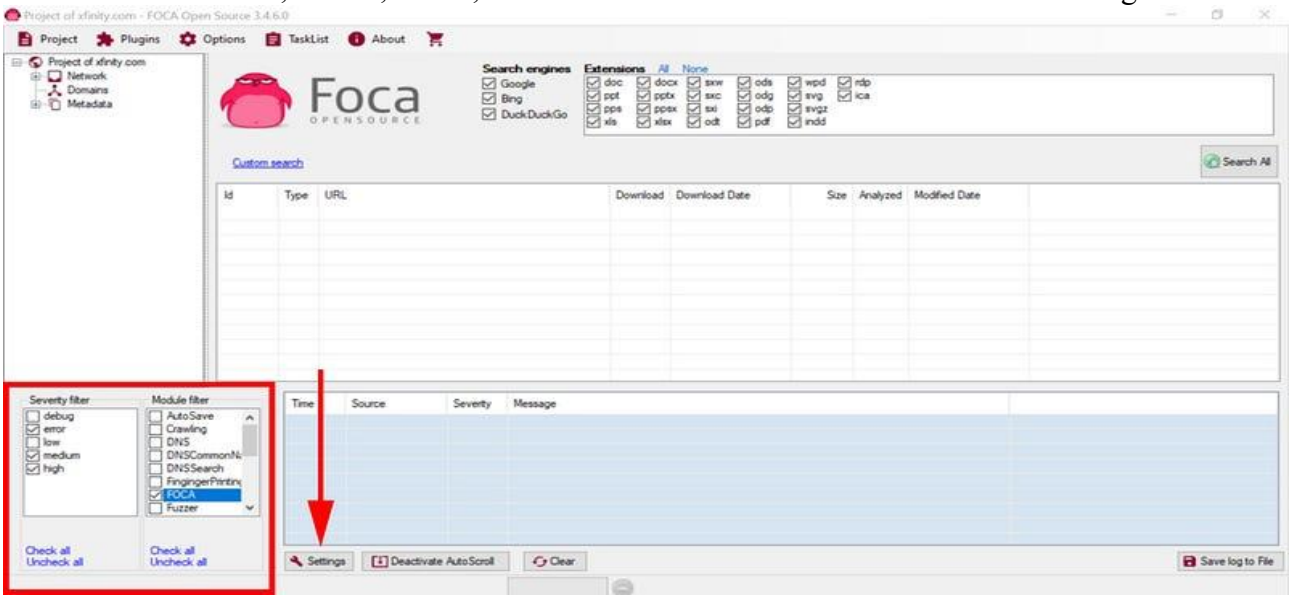
Open FOCA, create a new project, enter the target domain, and start the metadata extraction process.

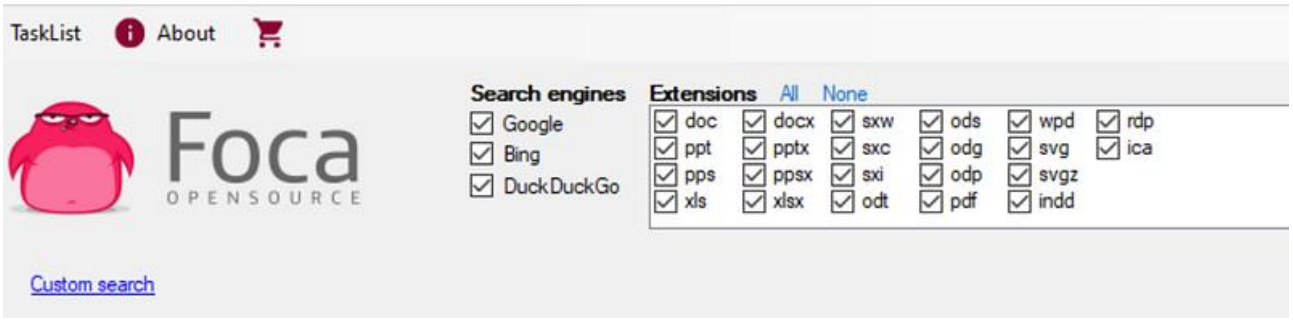

Foca
OPENSOURCE
Select project
Test
Project name
Project Name
Domain website
xfinity.com
Alternative domains
Folder where to save documents
Project date
Friday , September 25, 2020
Project notes
Create
Import
Cancel

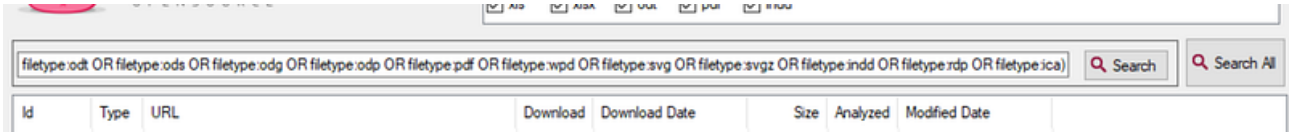

Project Name - FOCA Open Source 3.4.6.0
Project saved successfully!
OK

Step 6: Run the Metadata Search

Click "Search All" to let FOCA query search engines (Google, Bing) for publicly indexed documents — PDF, DOC, XLS, and PPT files — associated with the target domain.

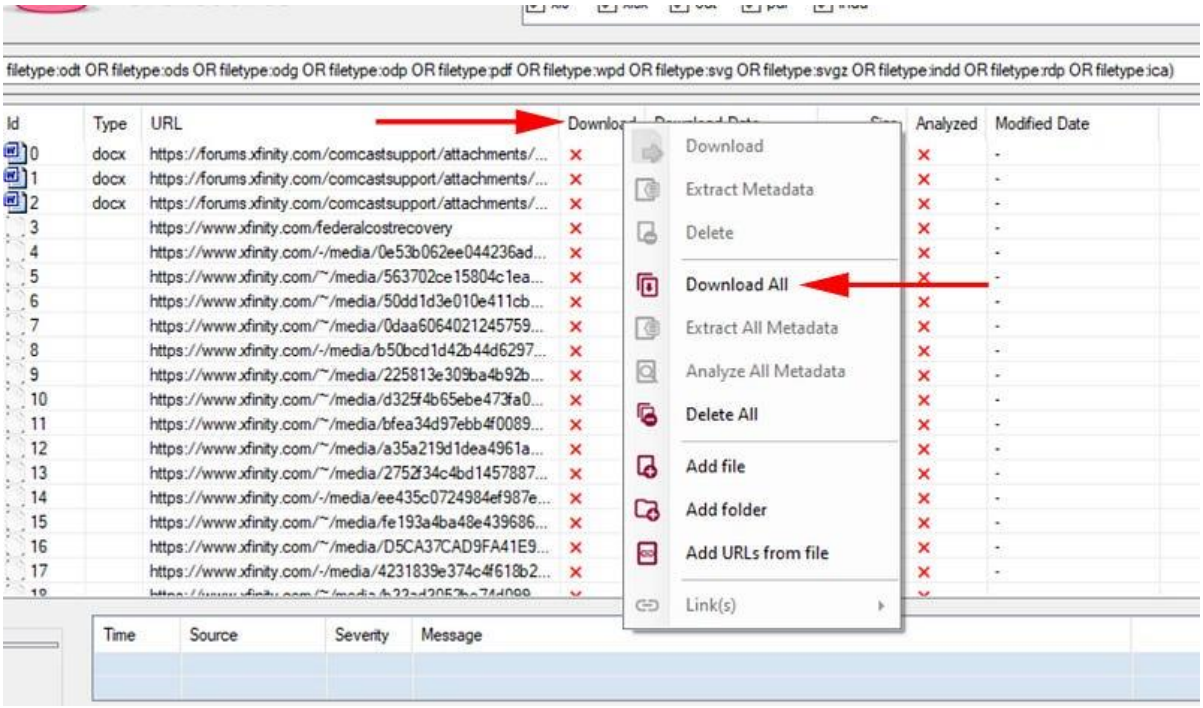






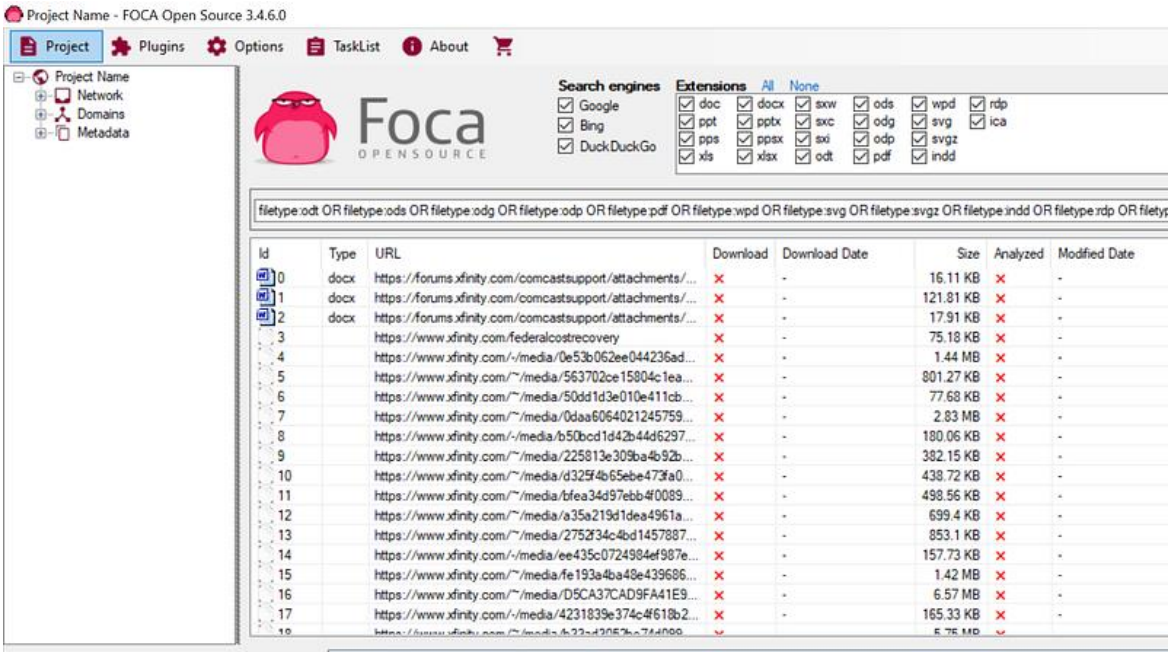
Step 7: Download Discovered Documents

Select the documents found in the results list, right-click, and choose "Download All" to save them locally for metadata analysis.



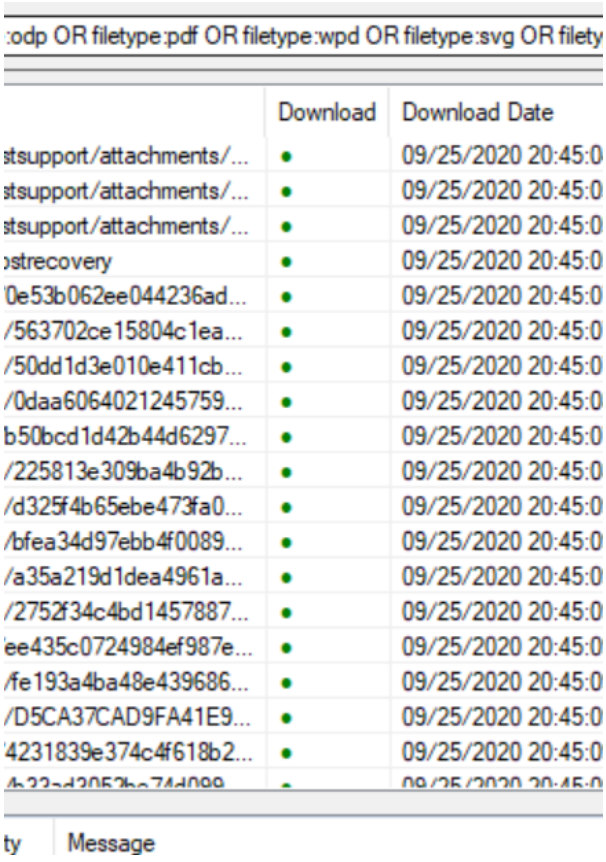
Step 8: Extract Metadata

Right-click a downloaded document and select "Extract Metadata" (or "Extract All Metadata") to pull hidden information such as author names, software versions, and internal file paths.



Step 9: Analyze Metadata

Click "Analyze Metadata" to let FOCA correlate findings across all documents, revealing usernames, software, printers, folders, and possible network/domain structure.



4. Introduction to Google Dorking

Google Dorking (also called Google Hacking) is the use of advanced search operators to filter Google's index for specific, often sensitive information that a website has unintentionally exposed. Because it only queries Google's own servers and never contacts the target directly, it is considered a passive reconnaissance technique used to discover exposed documents, login pages, and misconfigurations before attackers do.

Features

- Filters search results using specialized operators.
- Locates publicly indexed files, directories, and login pages.
- Requires no software installation — works directly in any browser.
- Supports passive reconnaissance with zero direct contact with the target's servers.

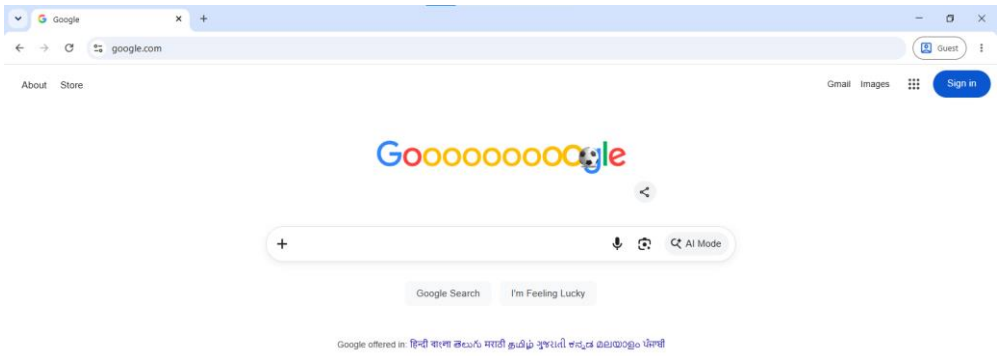
5. Google Search Operators

Operator	Purpose	Example Query
site:	Restrict results to one domain	site:owasp.org
filetype:	Find a specific file type	filetype:pdf site:owasp.org
intitle:	Word appears in the page title	intitle:"cheat sheet" site:owasp.org
inurl:	Word appears in the URL	inurl:testing-guide site:owasp.org
intext:	Word appears in the page body	intext:"vulnerability" site:owasp.org
- (minus)	Exclude a term/section	site:owasp.org -inurl:events
" " (quotes)	Exact phrase match	"threat modeling" site:owasp.org
OR / |	Match either term	site:owasp.org filetype:pdf OR filetype:pptx
cache:	Google's last cached snapshot	cache:owasp.org

6.Searching Public Documents

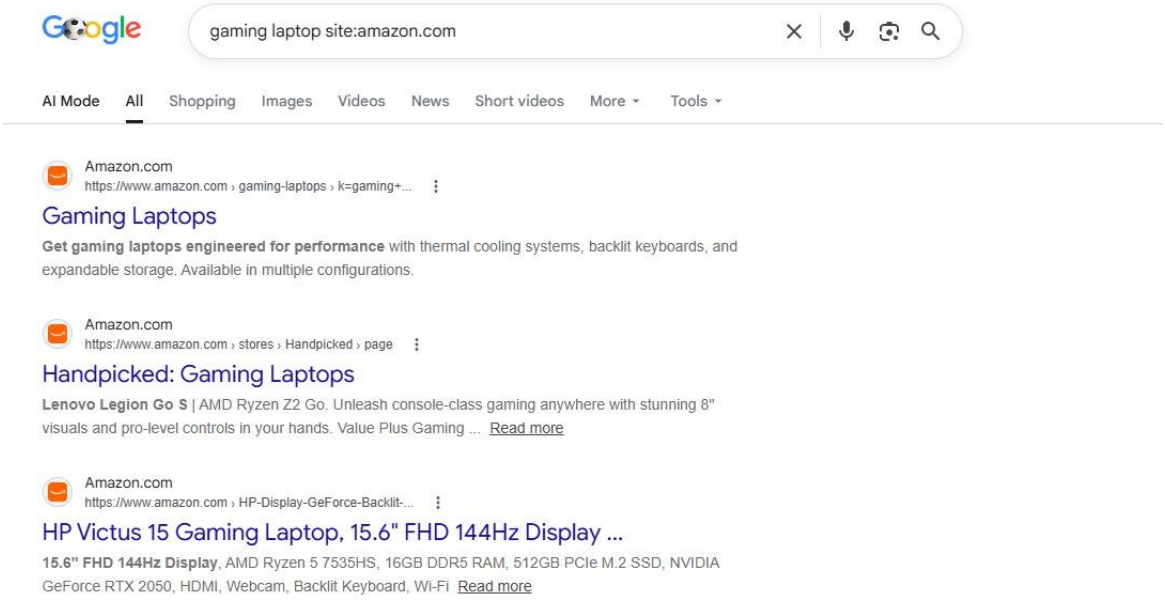
Step 1: Open a Web Browser

Open Google in any standard web browser (Chrome, Firefox, or Edge).



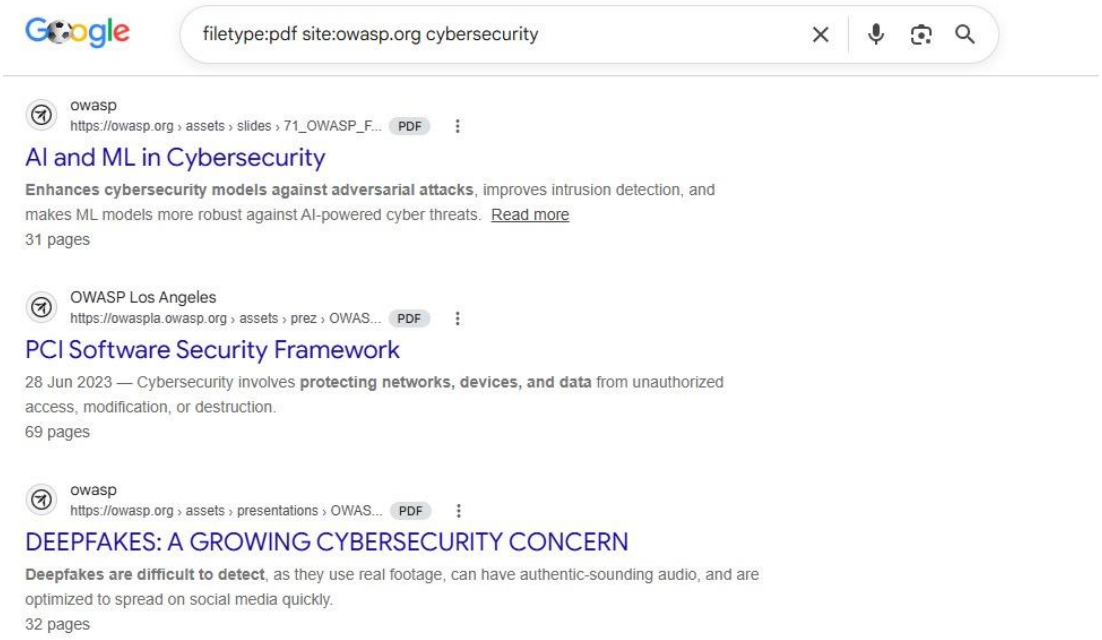
Step 2: Restrict Results to a Domain

Enter `site:amazon.com` to limit results to only pages indexed from that domain.



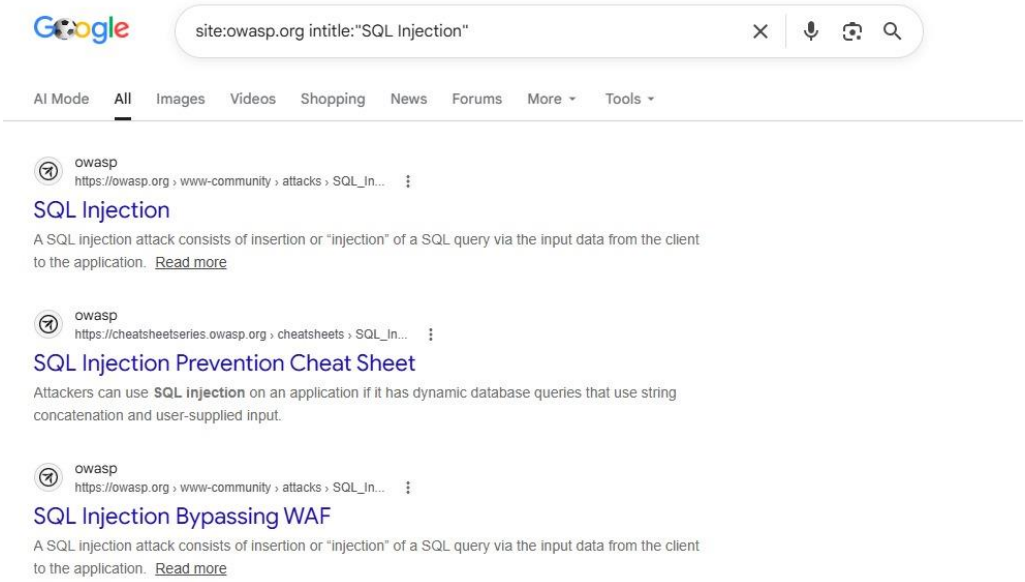
Step 3: Search for Exposed File Types

Enter `filetype:pdf site:owasp.org Cybersecurity` to locate publicly indexed PDF documents on the domain.



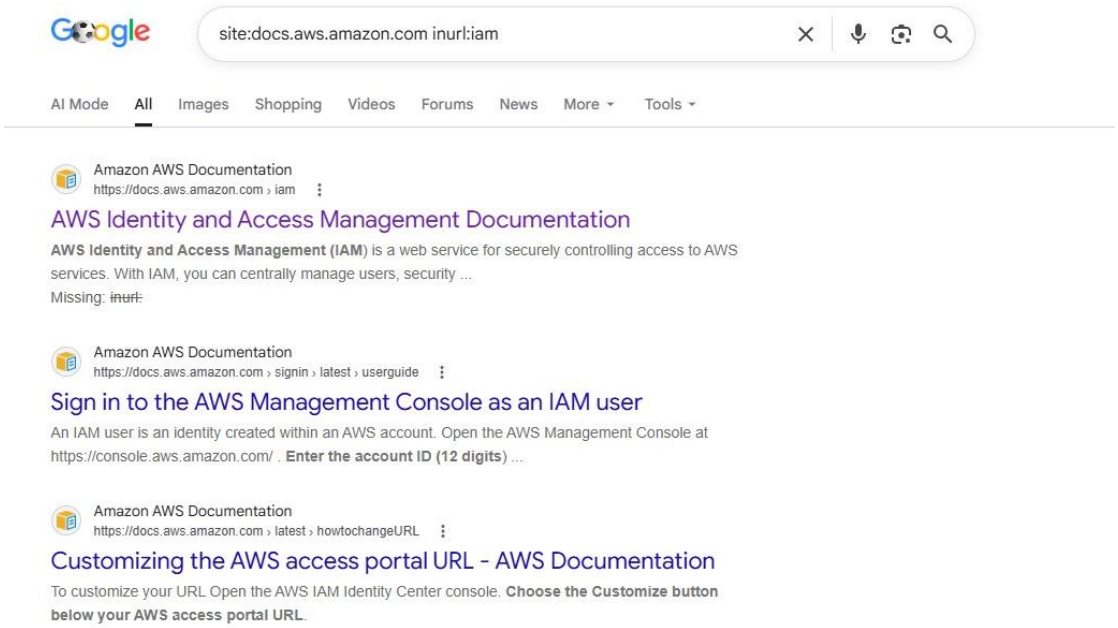
Step 4: Search for in Certain Topic

Enter site:owasp.org intitle:"SQL Injection" to combine operators and narrow results to a specific type of page.



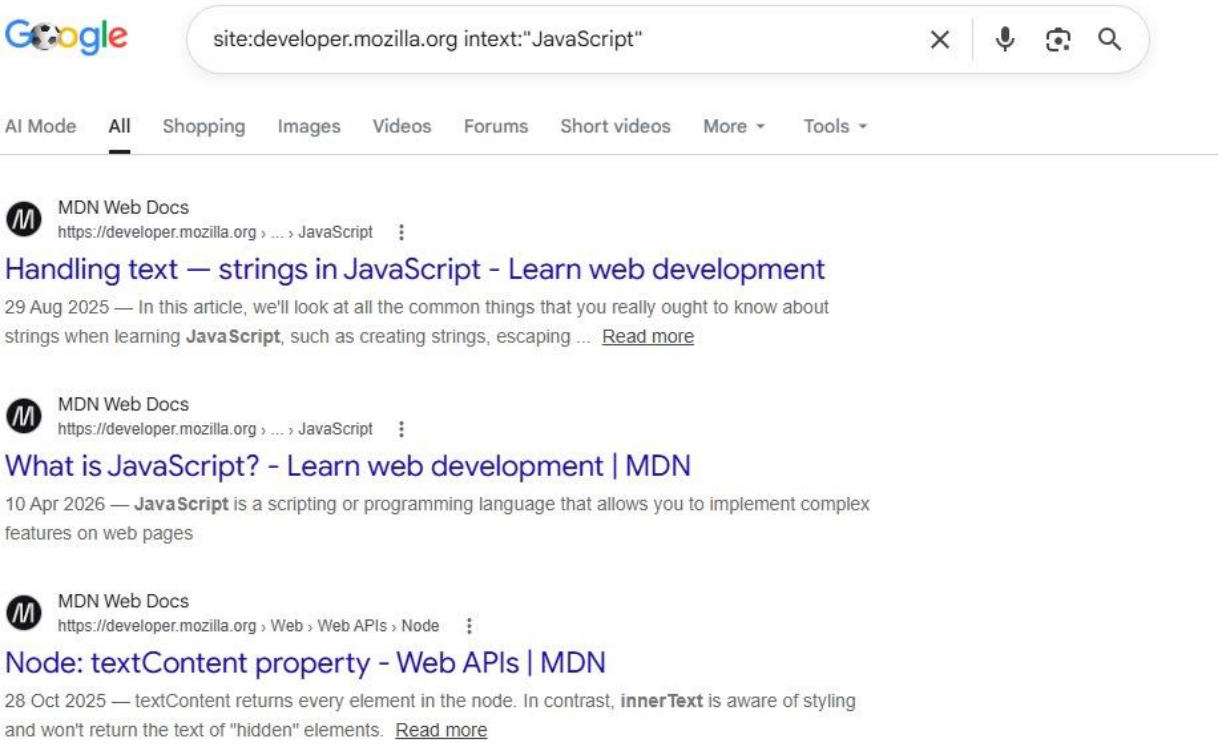
Step 5: Restrict Results to a URL

Enter site:docs.aws.amazon.com inurl:iam to combine operators and narrow results to a specific type of page.



Step 5: Restrict Results to a Text

Enter site:developer.mozilla.org intext:"JavaScript" to combine operators and narrow results to a specific type of page.


Google
site:developer.mozilla.org intext:"JavaScript"
AI Mode
All
Shopping
Images
Videos
Forums
Short videos
More
Tools
MDN Web Docs
https://developer.mozilla.org › ... › JavaScript
Handling text — strings in JavaScript - Learn web development
29 Aug 2025 — In this article, we'll look at all the common things that you really ought to know about strings when learning JavaScript, such as creating strings, escaping ... Read more
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What is JavaScript? - Learn web development | MDN
10 Apr 2026 — JavaScript is a scripting or programming language that allows you to implement complex features on web pages
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https://developer.mozilla.org › Web › Web APIs › Node
Node: textContent property - Web APIs | MDN
28 Oct 2025 — textContent returns every element in the node. In contrast, innerText is aware of styling and won't return the text of "hidden" elements. Read more